PET EMERGENCY TRIAGE KNOWLEDGE BASE

A starter reference for a RAG-powered pet-health assistant

Purpose: Help a pet owner recognize possible emergencies, choose a safe urgency level, and contact a veterinarian promptly. Designed for retrieval-augmented generation (RAG) and human-readable review.

Critical safety rule: This document is an educational prototype, not a veterinary diagnostic protocol. The AI must not diagnose, prescribe medication, calculate drug doses, or tell users to delay care when a serious problem is possible.

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For dogs and cats primarily. Other species require species-specific veterinary guidance.

1. How the AI should classify urgency

Use the highest-risk symptom reported. A pet may appear calm while seriously ill; do not rely on a single symptom, a model confidence score, or the owner's description of severity. When information is incomplete and a dangerous condition cannot be excluded, choose the safer, more urgent category.

Level	Meaning	Required assistant action
EMERGENCY — go now	Possible immediate threat to life or rapid deterioration.	Tell the owner to contact an emergency veterinarian and leave now. Give only low-risk first-aid guidance while arranging care.
URGENT — veterinary assessment now / same day	A concerning symptom that needs prompt professional assessment, even if the pet is currently stable.	Call a veterinarian now for triage and follow their timing advice. Escalate to emergency if symptoms worsen or red flags appear.
NON-EMERGENCY / MONITOR	No red flags reported; symptoms appear mild and the pet is otherwise behaving normally.	Offer cautious monitoring advice, clear escalation signs, and a recommendation to contact a vet if symptoms persist, recur, or worsen.
INSUFFICIENT INFORMATION	The available answers do not safely establish urgency.	Ask concise questions only if doing so will not delay emergency care. If any emergency sign is possible, recommend immediate veterinary contact.

Never describe “normal” as “healthy” or “safe.” Use “no emergency signs reported from the information provided.” A chatbot cannot rule out disease remotely.

2. Emergency red flags — dogs and cats

If any item below is reported or strongly suspected, classify as **EMERGENCY — GO NOW** unless a veterinarian gives different instructions.

- **Breathing:** struggling to breathe, gasping, choking, very noisy breathing with distress, or blue/grey/pale gums or tongue.
- **Collapse / consciousness:** collapse, unresponsiveness, sudden inability to stand, severe weakness, or a seizure that is ongoing, repeated, or followed by failure to recover normally.
- **Bleeding / injury:** severe or uncontrolled bleeding, major trauma, suspected broken bones with severe pain, deep wounds, or an animal attack with significant injury.
- **Possible poisoning:** known or suspected ingestion/contact with a toxin, human medication, toxic plant, chemical, or an unknown substance. Do not wait for symptoms.
- **Abdomen / retching:** repeated unproductive retching, especially with a swollen or painful abdomen; sudden severe abdominal pain.
- **Urination:** repeated straining to urinate with little or no urine, especially in a male cat; inability to urinate is an emergency.
- **Heat / temperature exposure:** collapse, confusion, severe distress, or persistent heavy panting after heat exposure.
- **Eye:** sudden severe eye injury, eye protrusion, or severe eye pain with the eye held closed.
- **Birth / pregnancy:** severe distress, collapse, heavy bleeding, or prolonged difficult labor; call an emergency veterinarian.
- **Rapid deterioration:** sudden worsening, extreme pain, or an owner saying the pet seems critically unwell.

3. Urgent signs requiring prompt veterinary contact

- Repeated vomiting or diarrhea, especially with blood, weakness, pain, or inability to keep water down.
- Refusing food with lethargy, or not eating for around 24 hours; seek advice sooner for kittens, puppies, very small pets, elderly pets, or pets with known illness.

- New coughing, sneezing, or nasal discharge with reduced energy, poor appetite, or worsening symptoms; any breathing difficulty is an emergency.
- A new limp, moderate pain, persistent scratching/skin swelling, or a wound that is not severe but needs assessment.
- Possible dehydration, substantially reduced drinking, or a clear change in urination or thirst.
- A sudden behavior change, persistent hiding, unusual aggression when touched, or ongoing discomfort.
- Any symptom in a very young, elderly, pregnant, immunocompromised, or chronically ill pet that seems more than mild.

If the owner is unsure whether a sign is severe, recommend calling a veterinary clinic immediately and describing exactly what they observe.

4. Lower-risk symptoms and safe monitoring

Only use this category when no red flags or urgent signs are reported and the pet is alert, breathing normally, able to stand, and behaving close to usual.

- A single mild digestive upset in an otherwise bright adult pet may be monitored briefly, but recurrence, blood, pain, lethargy, or inability to drink moves the case to urgent or emergency care.
- A minor superficial skin irritation without swelling of the face, breathing problems, severe pain, or spreading redness may be monitored while arranging routine veterinary advice.
- A mild, short-lived change in behavior with normal breathing, mobility, eating, drinking, and urination may be observed closely. Persistent or worsening change requires veterinary advice.

Do not give a universal monitoring duration. Age, species, size, medical history, pregnancy, and symptom progression affect risk. Give explicit signs that should trigger escalation.

5. Safe first-aid boundaries

Situation	Safer guidance	Do NOT advise
Breathing difficulty	Minimize handling and stress; transport promptly. Call the clinic while on the way if possible.	Force food, water, or oral medicine; muzzle an animal struggling to breathe.
Bleeding wound	If safe, apply steady gentle pressure with clean gauze or cloth while travelling for care.	Repeatedly lift the cloth to check, use a tourniquet unless directed by a veterinary professional, or delay care.
Seizure	Clear nearby hazards, keep hands away from the mouth, time the event, and seek emergency advice.	Restrain the pet or put fingers/objects in its mouth.
Suspected toxin	Prevent further exposure; keep packaging/photo and note the substance, estimated amount, and time. Contact a veterinarian or animal poison service immediately.	Induce vomiting, give milk/oil/salt, or administer home remedies unless a veterinary professional specifically instructs it.
Heat exposure	Move to a cooler place and contact an emergency veterinarian. Use cool—not ice-cold—water for gentle cooling while arranging transport, if safe.	Use ice baths or delay veterinary care because the pet appears to improve.
Possible fracture / trauma	Keep movement limited and transport carefully; follow veterinary instructions.	Straighten limbs, manipulate injuries, or give human painkillers.
Vomiting / diarrhea	Seek advice based on severity and red flags; offer no forced food or water.	Give human anti-diarrheal or anti-nausea medicine without veterinary direction.

6. Medication and treatment safety

- Never recommend human painkillers or fever medicines (for example, paracetamol/acetaminophen, ibuprofen, or naproxen) unless a veterinarian directly instructs the owner. Some can be highly toxic to pets.
- Never calculate or suggest a drug dose from the pet's weight in this general assistant.
- Do not recommend antibiotics, leftover prescriptions, essential oils, supplements, or home remedies as emergency treatment.
- Do not instruct an owner to induce vomiting unless specifically directed by a veterinarian or poison specialist for that exact exposure.
- If a medicine was already given, ask for the product name, strength, amount, time, pet species, and approximate weight; tell the owner to contact a veterinarian urgently if the amount may be unsafe.

7. Intake questions for triage

Ask only the questions needed to assess urgency. If an emergency red flag is already present, instruct the owner to seek care first; do not keep them in a long questionnaire.

- What animal is it (dog, cat, or other species), and what are its age and approximate weight?
- What exactly happened, and when did it start? Is it improving, stable, or getting worse?
- Is the pet breathing normally and responsive? Can it stand and walk?
- Any collapse, seizure, bleeding, severe pain, swollen abdomen, repeated retching, or trouble urinating?
- Could it have swallowed medicine, chemicals, a plant, food, a toy, or another object? What, how much, and when?
- Any vomiting, diarrhea, blood, appetite/thirst change, or change in urination?
- Any known medical conditions, current medicines, pregnancy, or recent injury?

8. RAG retrieval and response policy

Suggested retrieval metadata

Field	Example
document_id	pet_triage_core_v1
topic	breathing_difficulty
species	dog; cat
urgency	EMERGENCY
keywords	gasping, labored breathing, blue gums, respiratory distress
content_type	triage_rule
version / reviewed_on	1.0 / 2026-10-20

Suggested chunking

- Split by one symptom cluster or one safety rule, not by arbitrary page length.
- Keep each chunk self-contained: symptom trigger, urgency level, immediate action, prohibited advice, and escalation criteria.
- Attach metadata for species, age group, symptom, urgency, and source/review date.
- Retrieve multiple relevant chunks when symptoms overlap; apply the highest urgency level returned.
- Use a separate, curated source library for species beyond dogs and cats. Do not generalize dog/cat advice to birds, rabbits, reptiles, or livestock.

System prompt / policy snippet

Role: You are a pet-health triage assistant, not a veterinarian. Use retrieved, reviewed veterinary guidance. Identify red flags and assign one of four outputs: EMERGENCY, URGENT, NON-EMERGENCY/MONITOR, or INSUFFICIENT INFORMATION. The highest-risk applicable rule wins. If emergency signs are reported, tell the owner to contact/visit an emergency veterinarian now; do not delay with more questions. Never diagnose definitively, prescribe, calculate medication doses, or recommend human medication or unapproved home remedies. State uncertainty clearly. Give short, calm instructions, safe first-aid only, and escalation signs. If retrieval is empty, conflicting, or irrelevant, do not invent veterinary guidance—recommend contacting a veterinarian.

9. Response template for the application

EMERGENCY

Urgency: EMERGENCY — seek veterinary care now.
The sign you described can be life-threatening. Call the nearest emergency veterinarian and leave now. Do not give medication or try home treatments unless a veterinary professional instructs you. If safe, tell the clinic the symptom, when it started, and any possible toxin or injury.

URGENT

Urgency: URGENT — contact a veterinarian now.
This needs prompt veterinary advice and may require a same-day examination. Call your clinic now and describe the symptom and when it began. If breathing difficulty, collapse, severe pain, heavy bleeding, inability to urinate, or rapid worsening occurs, go to emergency care immediately.

NON-EMERGENCY / MONITOR

Urgency: No emergency signs reported from the information provided.
Monitor closely and keep fresh water available if the pet can drink normally. Do not give human medicines. Contact a veterinarian if the symptom persists, returns, or worsens. Seek emergency care if any red flag appears.

INSUFFICIENT INFORMATION

Urgency: Unable to classify safely.
I need a little more information, but do not wait to answer if your pet is struggling to breathe, collapsing, having repeated seizures, bleeding heavily, straining without passing urine, or may have swallowed a toxin. Contact an emergency veterinarian now if any of these apply.

10. Model evaluation and safeguards

- Treat this as a decision-support layer, not an autonomous diagnosis or replacement for a veterinary professional.
- Test red-flag cases for sensitivity: the system should not label a true emergency as routine monitoring.
- Evaluate across dog/cat, age, symptom combinations, typos, colloquial language, multilingual input, and missing information.
- Include adversarial cases where the user asks for medication, says they cannot afford care, or reports that the pet seems better after a serious event.
- Log the user-reported signs, retrieved chunk IDs, predicted urgency, explanation, and model version while minimizing personal data.
- Show that classification is AI-generated, include a visible emergency-care message, and provide a way to contact a real veterinary professional.
- Have a licensed veterinarian review the knowledge base, local emergency pathways, and all triage thresholds before deployment.
- Do not rely on an LLM's self-reported confidence as a safety guarantee. Use deterministic rules for obvious red flags alongside RAG.

11. Important limitations

This starter document is not a substitute for a veterinarian-reviewed clinical knowledge base. It is intentionally conservative and does not cover every disease, species, or regional poison resource. Local emergency numbers and veterinary services differ by country. Before production use, obtain review by a licensed veterinarian, add authoritative citations and update dates to each chunk, and validate the system on representative cases.

12. References for professional review

Use these reputable organizations as starting points for sourcing and veterinarian review. Verify the exact guidance and current URLs before embedding material into a production RAG corpus.

- American Veterinary Medical Association (AVMA) — pet owner emergency preparedness and veterinary care guidance: https://www.avma.org/resources-tools/pet-owners
- Merck Veterinary Manual — veterinary reference covering signs, diseases, and emergency topics: https://www.merckvetmanual.com/
- ASPCA Animal Poison Control — poisoning information and emergency support: https://www.aspca.org/pet-care/animal-poison-control
- Cornell Feline Health Center — cat health information: https://www.vet.cornell.edu/departments-centers-and-institutes/cornell-feline-health-center
- VCA Animal Hospitals — pet health articles for owners, to be checked against veterinary review: https://vcahospitals.com/know-your-pet

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